Van Nguyen
CET479 – Network Administration

Project 6 – Postfix and SMTP

Contents

Background

To install and configure a minimally functioning Postfix email server and send/retrieve a test email.

Objectives

1. Install postfix package.
2. Edit the Postfix configuration file for your unique implementation.
3. Configure DNS for local name resolution.
4. Install Telnet package.
5. Use Telnet to talk to the Postfix server and send a test message.
6. Install mailx mail client and read test message.

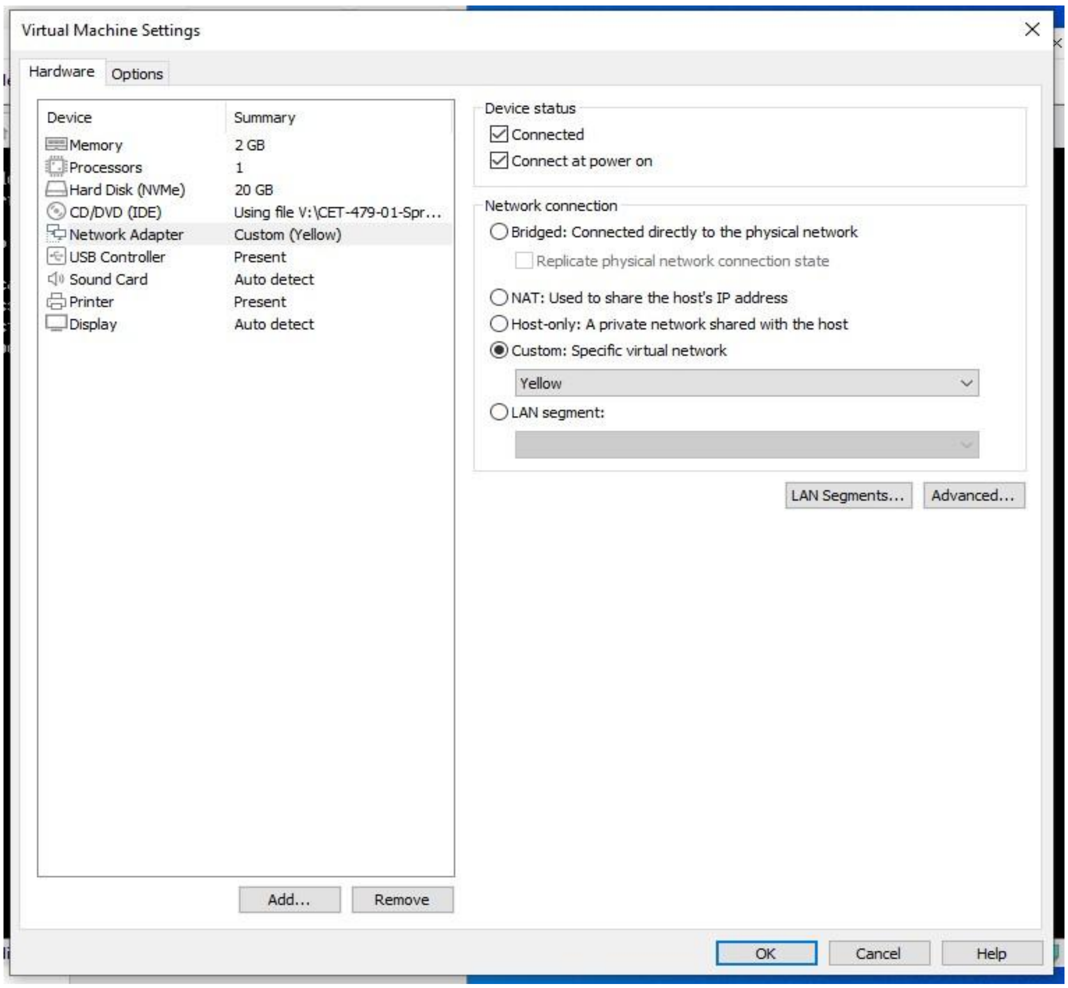
Prerequisites

- A hypervisor with a guest `running Fedora Server 35` or above
- A minimal network setup
- A local Fedora account setup to use sudo or root password for use with `su` command.

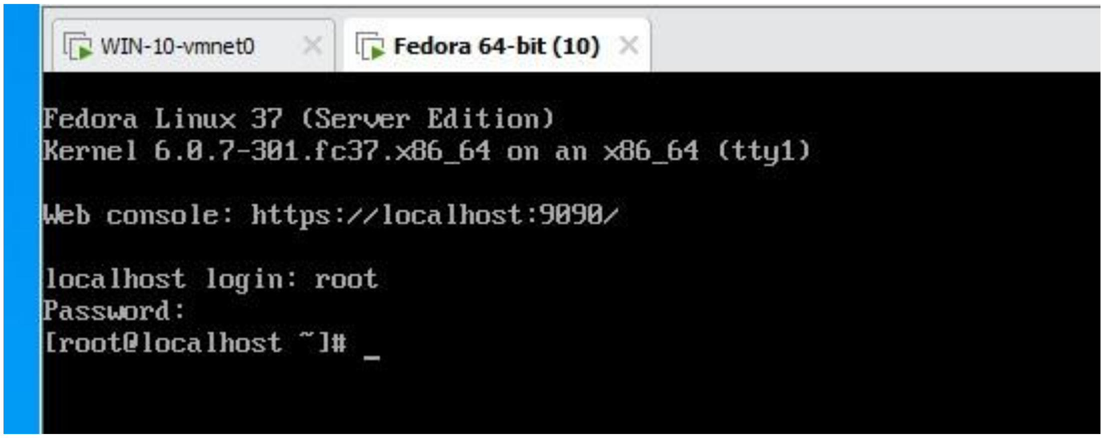
- A windows client (usually the local hypervisor host)

Connect your VM to the yellow network

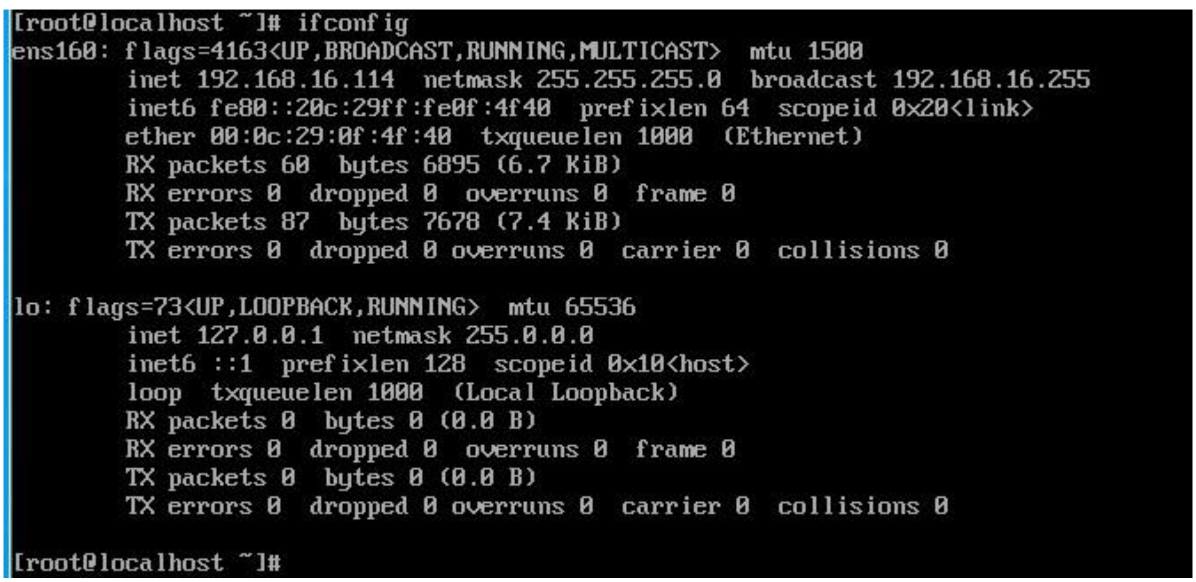
1. Before you start your VM, go to the VM settings and change the network configuration from **NAT** to **yellow.**



2. Start your VM and log in as **root**.



3. Type **ifconfig**. Notice that the IP address is not automatically assigned to your VM.



4. Type the following commands:

 # sudo nmcli connection modify ens160 ipv4.method auto # sudo nmcli connection down ens160; sudo nmcli connection up ens160

```
[root@localhost ~]# sudo nmcli connection modify ens160 ipv4.method auto
[root@localhost ~]# sudo nmcli connection down ens160; sudo nmcli connection up ens160
Error: 'ens160' is not an active connection.
Error: no active connection provided.
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnecti
on/1)
[root@localhost ~]#
```

5. Type **ifconfig again** to check if the IP address was now assigned. Add screenshot.

```
[root@localhost ~]# ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.16.114  netmask 255.255.255.0  broadcast 192.168.16.255
        inet6 fe80::20c:29ff:fe0f:4f40  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:0f:4f:40  txqueuelen 1000  (Ethernet)
        RX packets 274  bytes 26742 (26.1 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 38  bytes 3283 (3.2 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[root@localhost ~]#
```

Configure DNS

1. Configure local **DNS** name resolution on the Windows host and `Fedora guest.`

2. Setup Windows Client for Local Name Resolution

3. Open cmd and browse to `c:\windows\system32\drivers\etc`
4. Edit hosts with notepad.exe or similar tool and add the following lines where

   xxx.xxx.xxx.xxx = your Fedora server IP address:

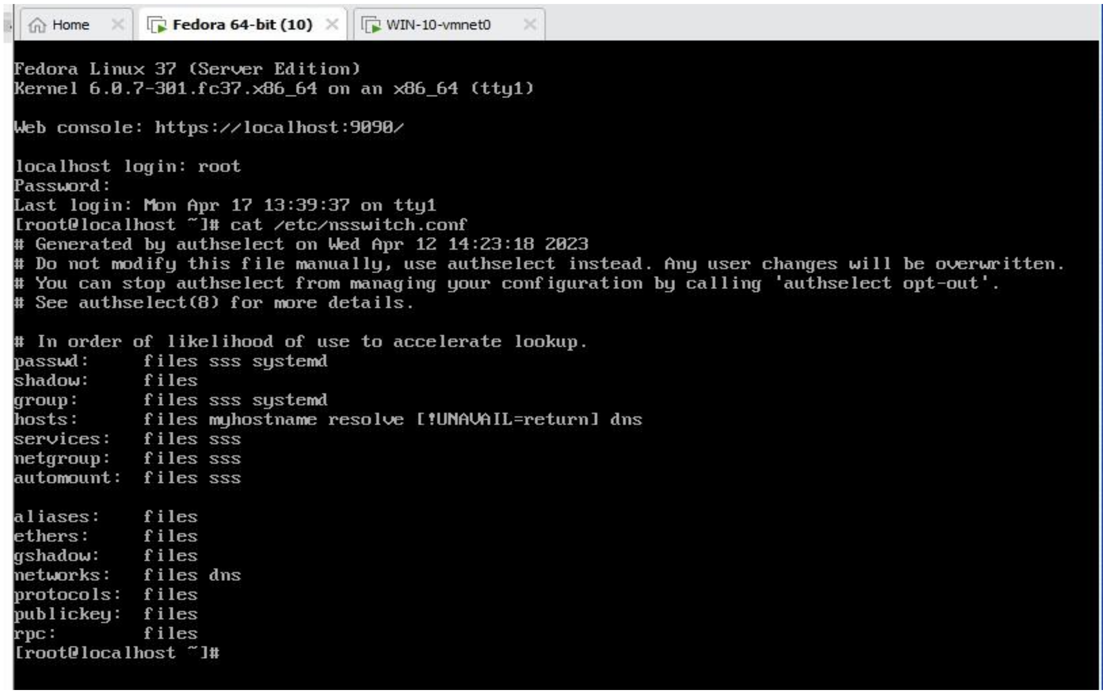
   xxx.xxx.xxx.xxx station.domain.networking.local xxx.xxx.xxx.xxx

   www.domain.networking.local

5. Setup Fedora Server for Local Name Resolution

6. Check `cat /etc/nsswitch.conf` to ensure the following is configured: hosts:

   files dns localhost

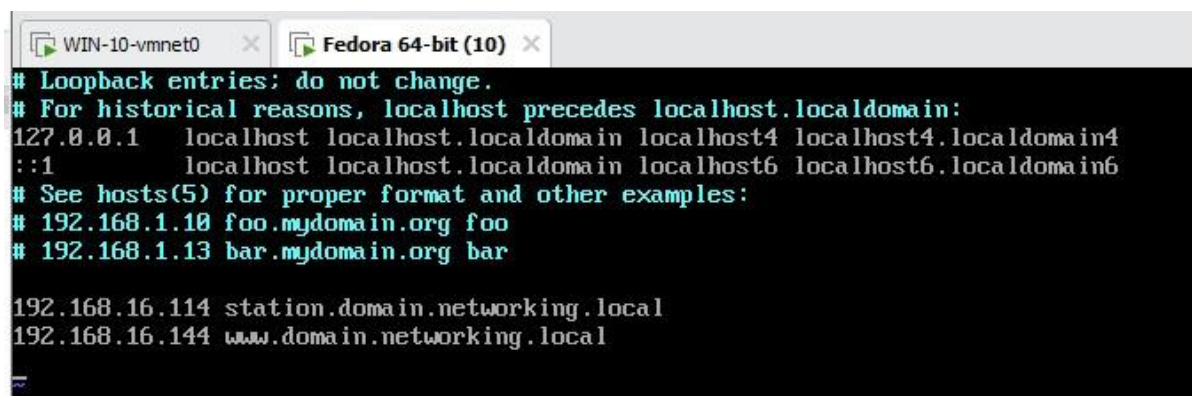



```
Fedora Linux 37 (Server Edition)
Kernel 6.0.7-301.fc37.x86_64 on an x86_64 (tty1)

Web console: https://localhost:9090/

localhost login: root
Password:
Last login: Mon Apr 17 13:39:37 on tty1
[root@localhost ~]# cat /etc/nsswitch.conf
# Generated by authselect on Wed Apr 12 14:23:18 2023
# Do not modify this file manually, use authselect instead. Any user changes will be overwritten.
# You can stop authselect from managing your configuration by calling 'authselect opt-out'.
# See authselect(8) for more details.

# In order of likelihood of use to accelerate lookup.
passwd:     files sss systemd
shadow:     files
group:      files sss systemd
hosts:      files myhostname resolve [!UNAVAIL=return] dns
services:   files sss
netgroup:   files sss
automount:  files sss

aliases:    files
ethers:     files
gshadow:    files
networks:   files dns
protocols:  files
publickey:  files
rpc:        files
[root@localhost ~]#
```

7. **Edit /etc/hosts** and add the following lines where xxx.xxx.xxx.xxx = your Fedora server IP address

8. **vi /etc/hosts**

xxx.xxx.xxx.xxx	station.domain.networking.local
xxx.xxx.xxx.xxx	www.domain.networking.local




```
# Loopback entries; do not change.
# For historical reasons, localhost precedes localhost.localdomain:
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
# See hosts(5) for proper format and other examples:
# 192.168.1.10 foo.mydomain.org foo
# 192.168.1.13 bar.mydomain.org bar

192.168.16.114 station.domain.networking.local
192.168.16.144 www.domain.networking.local
~
```

Telnet Installation

1. Install Telnet Server using the following command:

   ```
   yum install telnet-server
   ```

2. Start the Telnet Server:

   ```
   systemctl start telnet.socket
   ```

3. Install Telnet client using the following command:

   ```
   yum install telnet
   ```

```
Home    Fedora 64-bit (10)    WIN-10-vmnet0

Complete!
[root@localhost ~]# systemctl start telnet.socket
[root@localhost ~]# yum install telnet
Last metadata expiration check: 0:00:56 ago on Mon 17 Apr 2023 02:43:55 PM EDT.
Dependencies resolved.
================================================================================================
 Package              Architecture          Version                     Repository          Size
================================================================================================
Installing:
 telnet               x86_64                1:0.17-87.fc37              fedora              64 k

Transaction Summary
================================================================================================
Install  1 Package

Total download size: 64 k
Installed size: 122 k
Is this ok [y/N]: y
Downloading Packages:
telnet-0.17-87.fc37.x86_64.rpm                                    123 kB/s |  64 kB     00:00
------------------------------------------------------------------------------------------------
Total                                                              62 kB/s |  64 kB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                         1/1
  Installing       : telnet-1:0.17-87.fc37.x86_64                                            1/1
  Running scriptlet: telnet-1:0.17-87.fc37.x86_64                                            1/1
  Verifying        : telnet-1:0.17-87.fc37.x86_64                                            1/1

Installed:
  telnet-1:0.17-87.fc37.x86_64

Complete!
[root@localhost ~]#
```

4. Install Postfix using the following command:

   yum install postfix

```
Complete!
[root@localhost ~]# yum install postfix
Last metadata expiration check: 0:04:15 ago on Mon 17 Apr 2023 02:43:55 PM EDT.
Package postfix-2:3.7.3-1.fc37.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@localhost ~]# _
```

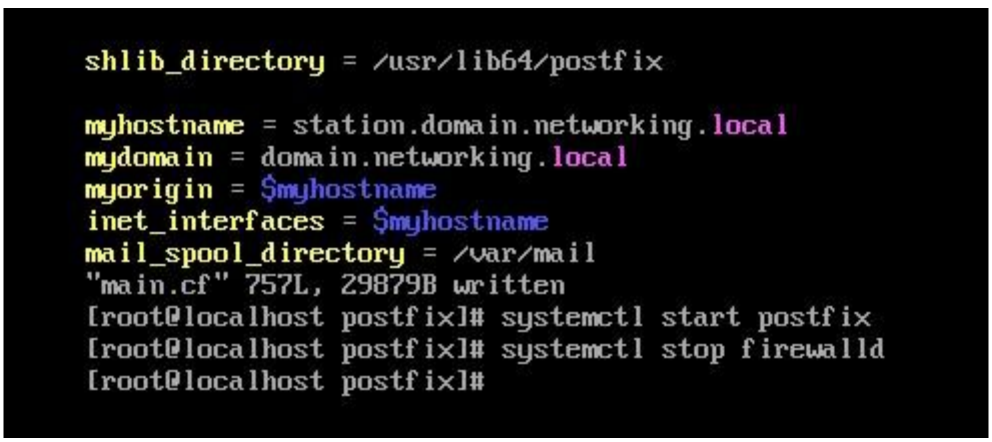
5. **cd** to **/etc/postfix** and **vi main.cf** and set the following configuration variables:

    myhostname = station.domain.networking.local
    mydomain = domain.networking.local myorigin
    = $myhostname inet_interfaces = $myhostname
    mail_spool_directory = /var/mail

6. Start the email server:

    systemctl start postfix

7. Stop the firewall service: systemctl stop firewalld

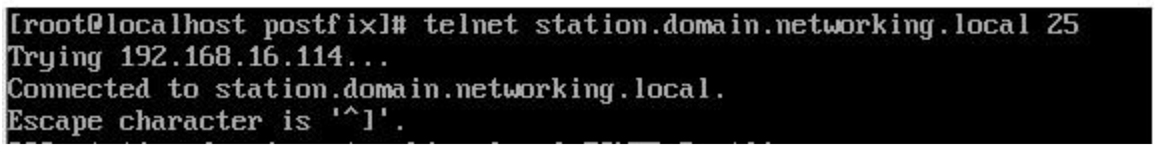


```
shlib_directory = /usr/lib64/postfix

myhostname = station.domain.networking.local
mydomain = domain.networking.local
myorigin = $myhostname
inet_interfaces = $myhostname
mail_spool_directory = /var/mail
"main.cf" 757L, 29879B written
[root@localhost postfix]# systemctl start postfix
[root@localhost postfix]# systemctl stop firewalld
[root@localhost postfix]#
```

Communicate with Email Server & Send Test Message

1. Use telnet to talk directly to the mail server (make sure to change default port from 23 to 25):

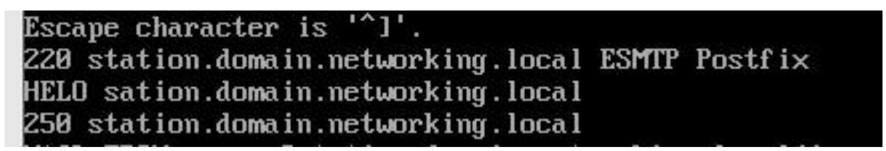
   telnet station.domain.networking.local 25



```
[root@localhost postfix]# telnet station.domain.networking.local 25
Trying 192.168.16.114...
Connected to station.domain.networking.local.
Escape character is '^]'.
```

2. Add the HELO command to tell the server which domain you are coming from:

   HELO station.domain.networking.local



```
Escape character is '^]'.
220 station.domain.networking.local ESMTP Postfix
HELO sation.domain.networking.local
250 station.domain.networking.local
```

3. Add the sender using the MAIL FROM command:

   MAIL FROM: <user>@station.domain.networking.local (in my case user would be riti@station.domain.networking.local)

4. Add the recipient using RCPT TO: <user>@station.domain.networking.local (in my case user would be riti@station.domain.networking.local).

   Note: In this example I'm sending myself a test message.

```
Connection closed by foreign host.
[root@localhost postfix]# telnet station.domain.networking.local 25
Trying 192.168.16.114...
Connected to station.domain.networking.local.
Escape character is '^]'.
220 station.domain.networking.local ESMTP Postfix
HELO station.domain.networking.local
250 station.domain.networking.local
MAIL FROM:van@station.domain.networking.local
250 2.1.0 Ok
RCPT TO:van@station.domain.networking.local
550 5.1.1 <van@station.domain.networking.local>: Recipient address rejected: User unknown in local r
ecipient table
_
```

5. Finally, we use the DATA command to add content to the message:

   Type DATA and press enter

   Type Subject: This is a test message and press enter

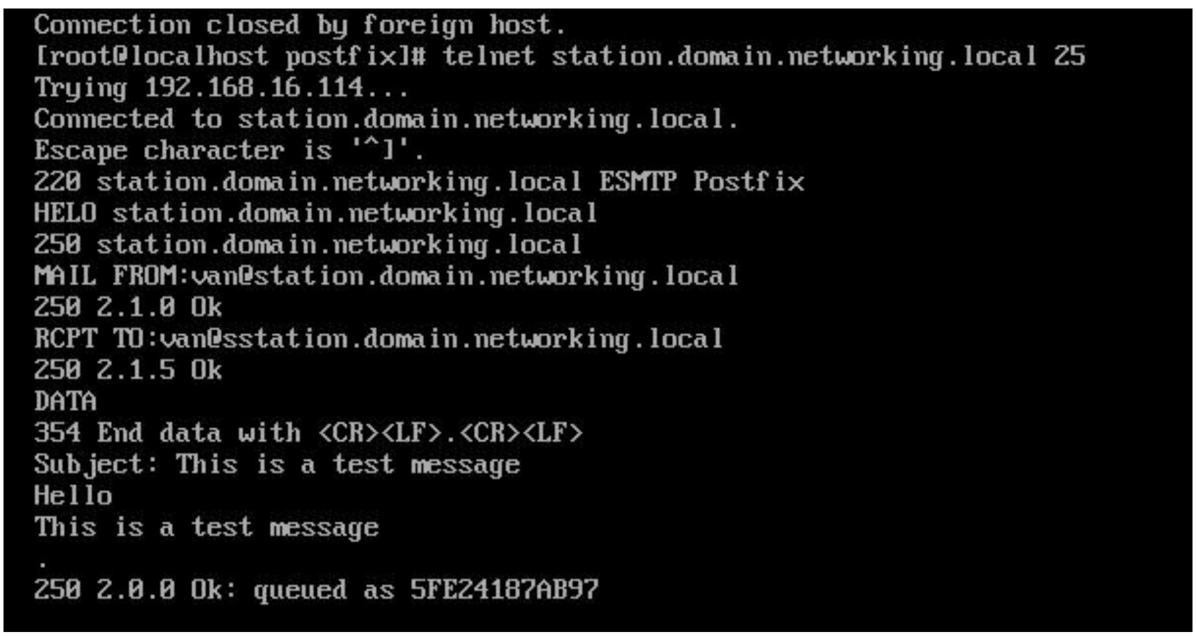
   Type Hello, and press enter

   Type This is a test message and press enter

   Type . and press enter (just a period)

   The . used above on a line by itself is used to finish the message body. The code response will notify you if the email was successful or not.

   Once done, use the quit command to close the session.

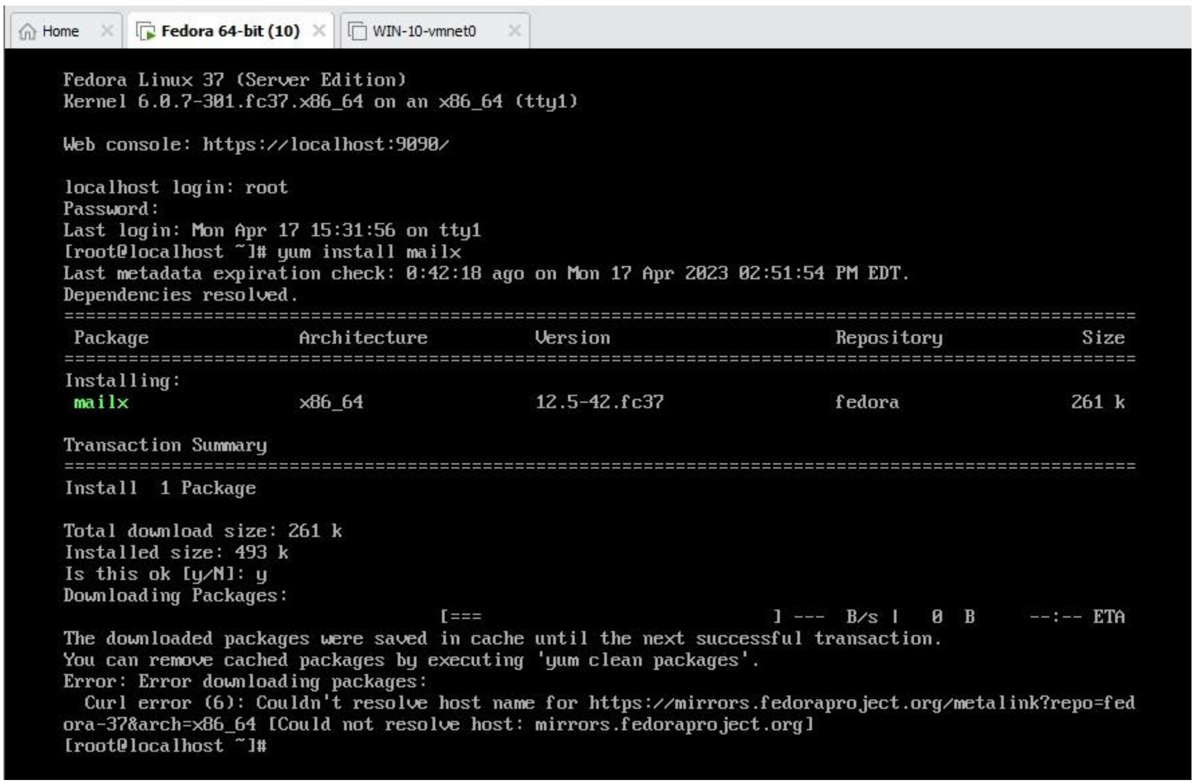


```
Connection closed by foreign host.
[root@localhost postfix]# telnet station.domain.networking.local 25
Trying 192.168.16.114...
Connected to station.domain.networking.local.
Escape character is '^]'.
220 station.domain.networking.local ESMTP Postfix
HELO station.domain.networking.local
250 station.domain.networking.local
MAIL FROM:van@station.domain.networking.local
250 2.1.0 Ok
RCPT TO:van@sstation.domain.networking.local
250 2.1.5 Ok
DATA
354 End data with <CR><LF>.<CR><LF>
Subject: This is a test message
Hello
This is a test message
.
250 2.0.0 Ok: queued as 5FE24187AB97
```

Read Test Message

1. On the Fedora server, login with your root credentials.

2. Install mail client:
   yum install mailx



3. Now log in to your non root account and complete the following steps:

4. Use the mail command with no arguments to display all messages in your mailbox:
   mail

The mail client will open and display a list of the email messages in the mailbox. Each is numbered. To read one, enter the number of the message.

5. Type quit to exit.

Submission

Use this document (or one of your choosing) to submit your screen shots and any other relevant data. At a minimum, you should show the commands run after logging in. Please ensure your submission includes your name, class, date, and reference to this lab.